1)

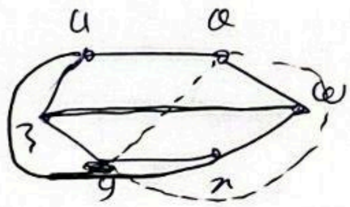
Planarity :

Theorem 1:

The graph K_5 and $K_{3,3}$ are non-planar.

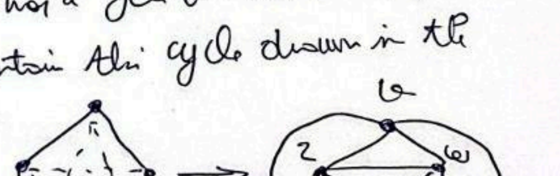
proof:

Suppose that first $K_{3,3}$ is planar, since $K_{3,3}$ has a cycle $u \to v \to w \to x \to y \to u$ of length 6, any plane drawing must contain this cycle drawn in the form of hexagon.



Now the edge wz must lie either wholly inside the hexagon or wholly outside it. We deal with the case in which wz lies inside the hexagon — the other case is similar. Since the edge ux must not cross the edge wz, it must lie outside the hexagon, it is then impossible to draw the edge vy, as it would cross either ux or wz.

. Now suppose K_5 is planar, since it has a cycle $v \to w \to x \to y \to z \to v$ of length 5. Any plane drawing must contain this cycle drawn in the form of pentagon.



Now, the edge wz must lie either wholly inside the pentagon or wholly outside it. We deal with the case in which wz lies inside the pentagon — the other case is similar. Since the edges vx and vy do not cross the edge wz must both lie outside the pentagon; but the edge xz cannot cross xz cannot cross edge vy and so must lie inside the pentagon, similarly wy must lie inside then wy and xz must cross.

Theorem 2 : (Kuratowski 1930)
a graph is planar if and only if no subgraph homomorphic to K_5 or $K_{3,3}$.

Theorem 3 :
a graph is planar if and only if ~~it~~ no subgraph contractible to K_5 or $K_{3,3}$.

Handshaking lemma for faces :
in any plane drawing of a planar graph,

$$\sum \deg(f_i) = 2\,|E(G)|.$$

proof :
in a plane drawing each edge has two sides (two incident faces)
. the edge may lie on the boundary of a single face (count 2)
. the edge lies on the boundary of face f_1 on one side, and a face f_2 on the other side.
then each edge contribute exactly 2 to the total degree.

$$\Rightarrow \sum \deg(f_i) = 2\,|E(G)|.$$

Theorem 4 (Euler 1750):
let G a plane drawing of a connected planar graph and let n, m and f resp the number of vertices, edges and faces of G. Then,
$$n - m + F = 2.$$
proof: by induction on the number of edges of G. If $m=0$, then $n = 1$, $f = 1$. The theorem is True in this case.
Now, suppose that the theorem holds for all graph with at most $m-1$ edges. let G be a graph with m edges, if G is a tree then $m = n - 1$, $f = 1$, so that ~~n~~ $n - m + f = 2$ as required.
if G is not a tree. let e be an edge in some cycle of G then $G-e$ is a connected plane graph with n vertices, $m-1$ edges, $f-1$ faces, so that $n-(m-1)+(f-1) = 2$ by induction hypothesis, adding the removed edge gives $n - m + F = 2$ as required.
Corollary 1: i) if G is a connected simple planar graph with $n (\geq 3)$ vertices and m edges, then: $m \leq 3n-6$
ii) - if in addition G has no triangle then $m \leq 2n-4$.
proof: i) Since each face is bounded by at least three edges, it follows on counting up the edges around each face, that: $\underline{3f \leq 2m}$ (the factor 2 appears since each edge bounds two faces), we obtain the ~~following~~ required formula by replacing $f = 2 - n + m$.
ii) this follows in similar way by replacing $3f \leq 2m$ ~~by~~ with $4f \leq 2m$.

Theorem 1:

$K_{3,3}$ and K_5, are non - planar

$K_{3,3}$: $n=6$, $m=9$ (no triangles)

$2n-4=8 < m=9$

$\Rightarrow K_{3,3}$ is non-planar

K_5 : $n=5$, $m=10$

$3n-6=9 < m=10$

$\Rightarrow K_5$ is non planar.

Corollary 2: let G be a simple connected graph, then G contains a vertex of degree 5 or less.

<u>proof</u>: $d_i \geq 6$.

$\sum d_i \geq 6n$

$2m \geq 6n$ (contradiction with $m \leq 3n-6$)

$m \geq 3n > 3n-6$